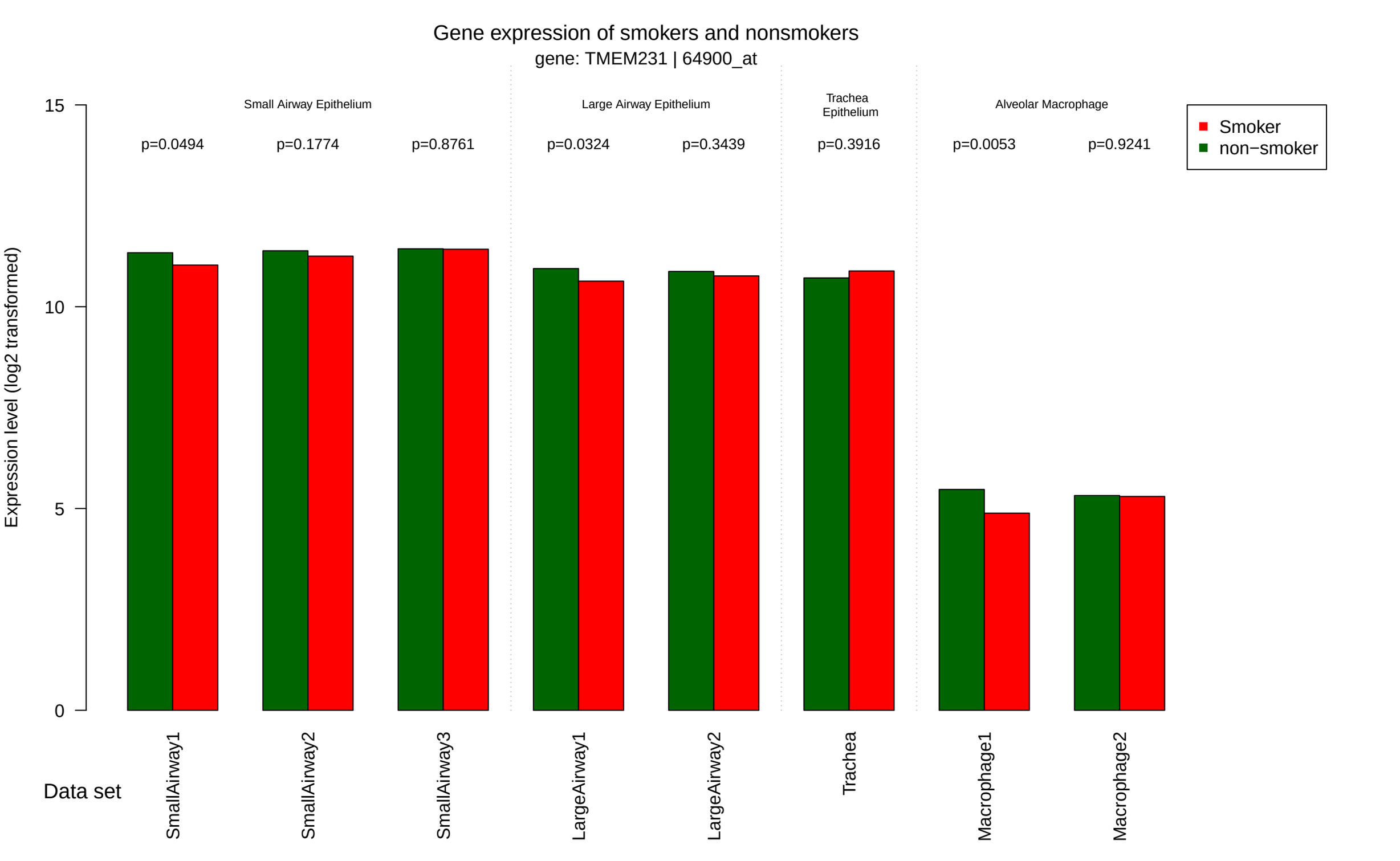
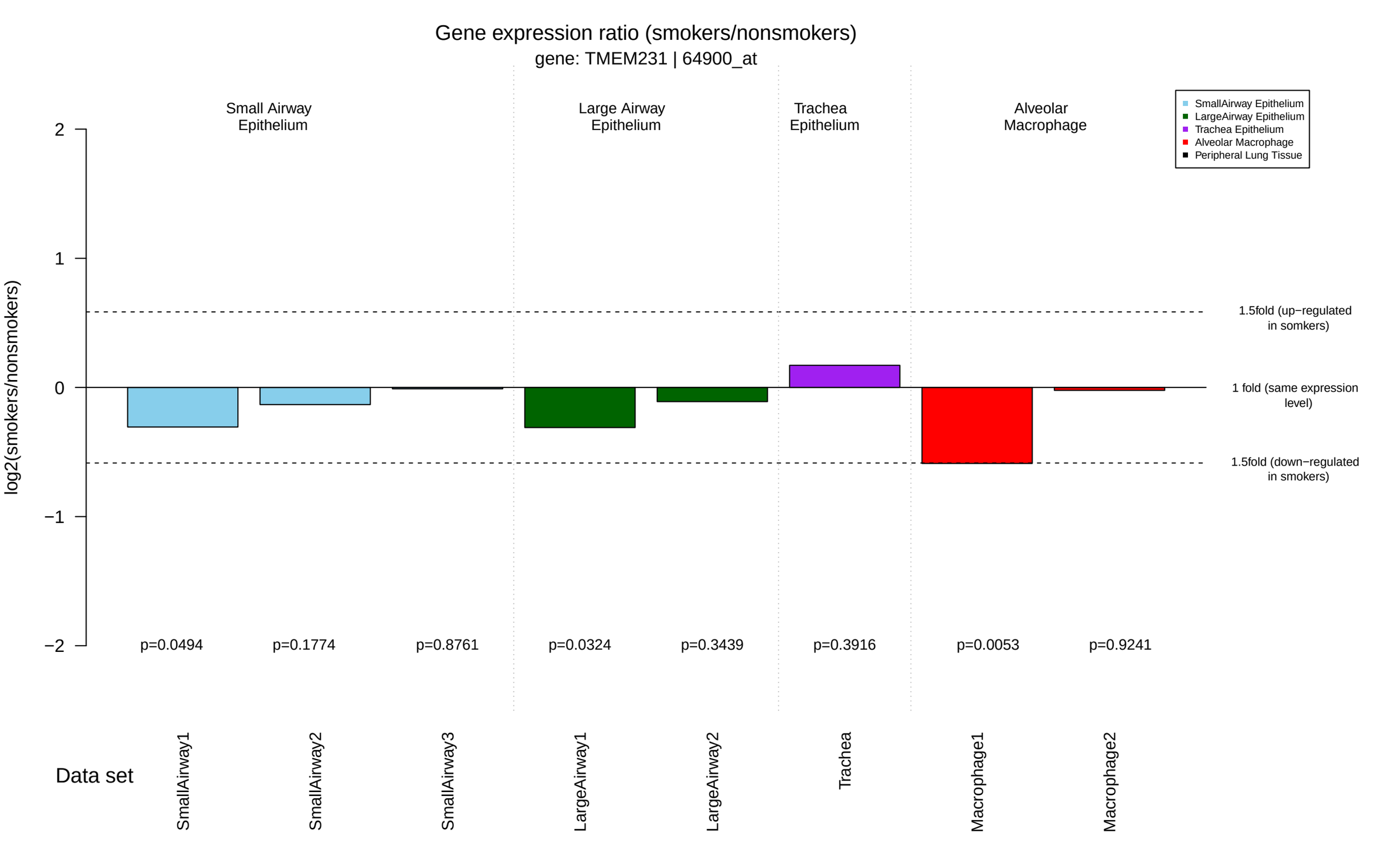
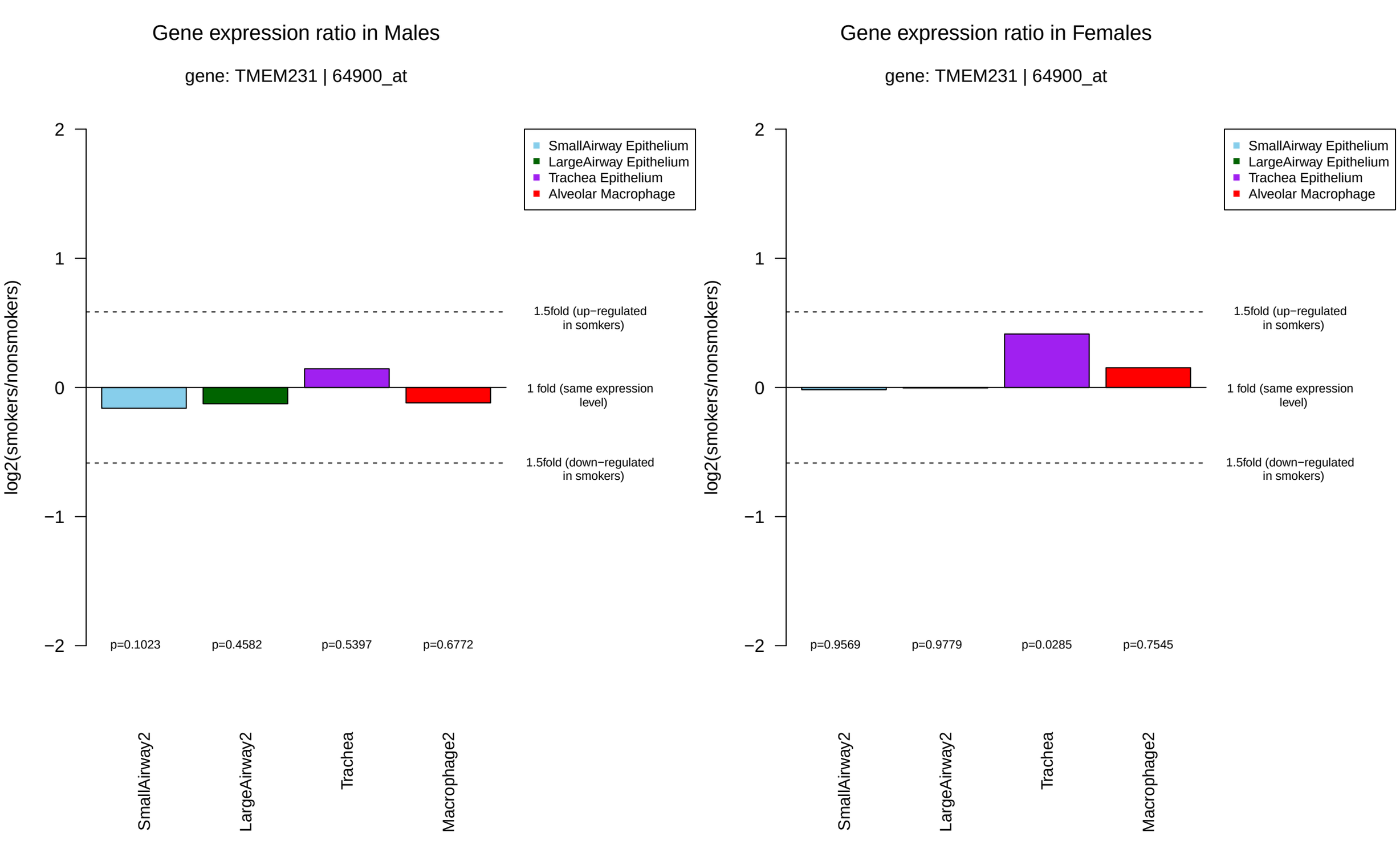
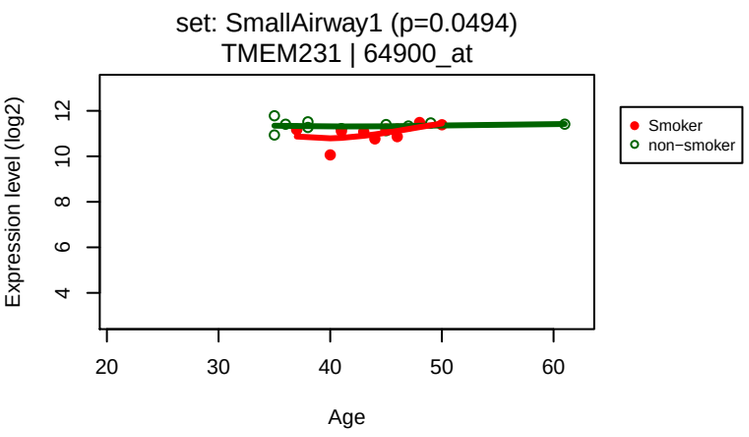
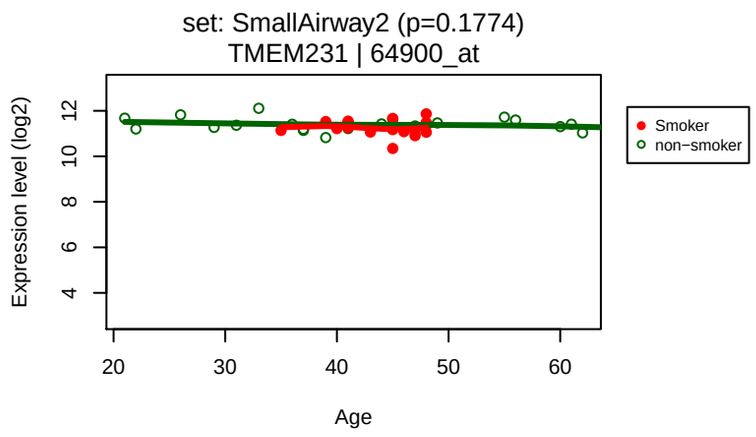
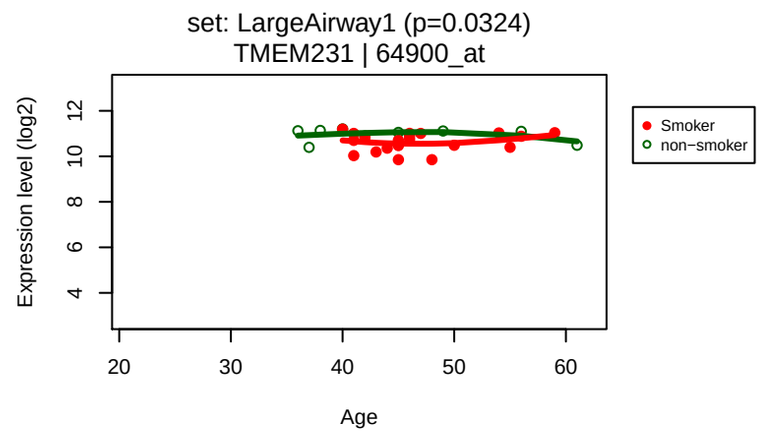

Gene expression of smokers and nonsmokers
gene: TMEM231 | 64900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0494
p=0.1774
p=0.8761
p=0.0324
p=0.3439
p=0.3916
p=0.0053
p=0.9241
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

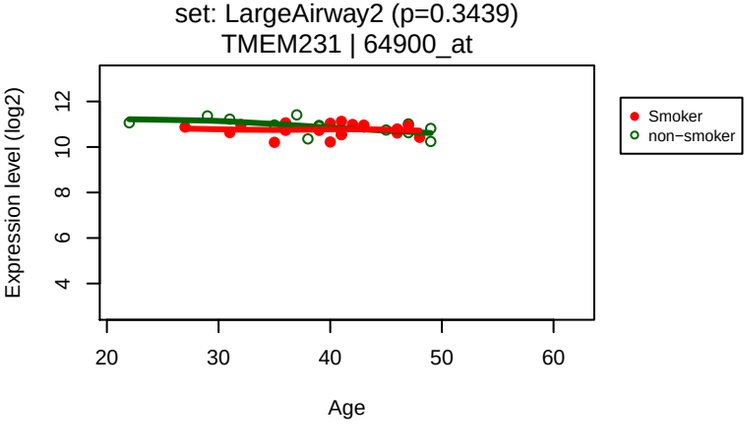

Gene expression ratio (smokers/nonsmokers)
gene: TMEM231 | 64900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0494
p=0.1774
p=0.8761
p=0.0324
p=0.3439
p=0.3916
p=0.0053
p=0.9241
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

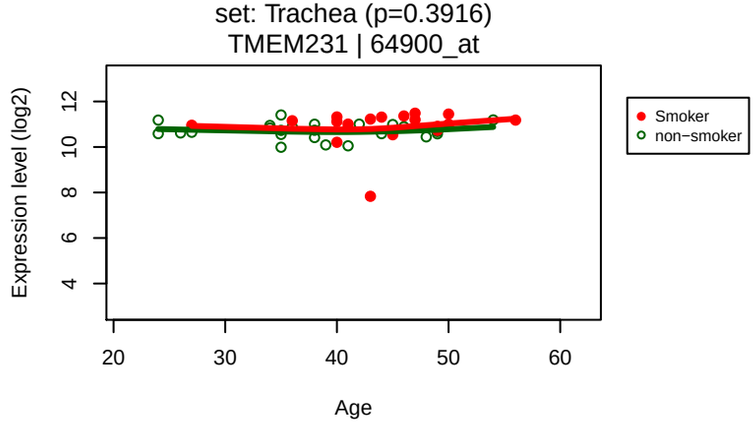

Gene expression ratio in Males
gene: TMEM231 | 64900_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1023
p=0.4582
p=0.5397
p=0.6772
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM231 | 64900_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9569
p=0.9779
p=0.0285
p=0.7545
SmallAirway2
LargeAirway2
Trachea
Macrophage2

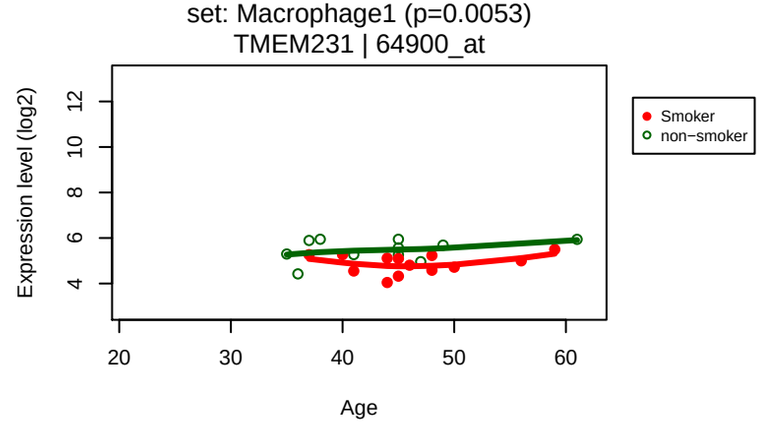

set: SmallAirway1 (p=0.0494)
TMEM231 | 64900_at
Expression level (log2)
Age
Smoker
non−smoker

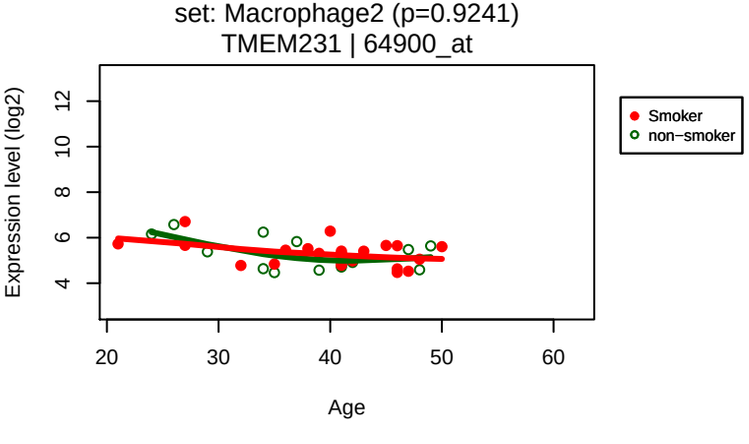

set: SmallAirway2 (p=0.1774)
TMEM231 | 64900_at
Expression level (log2)
Age
Smoker
non−smoker

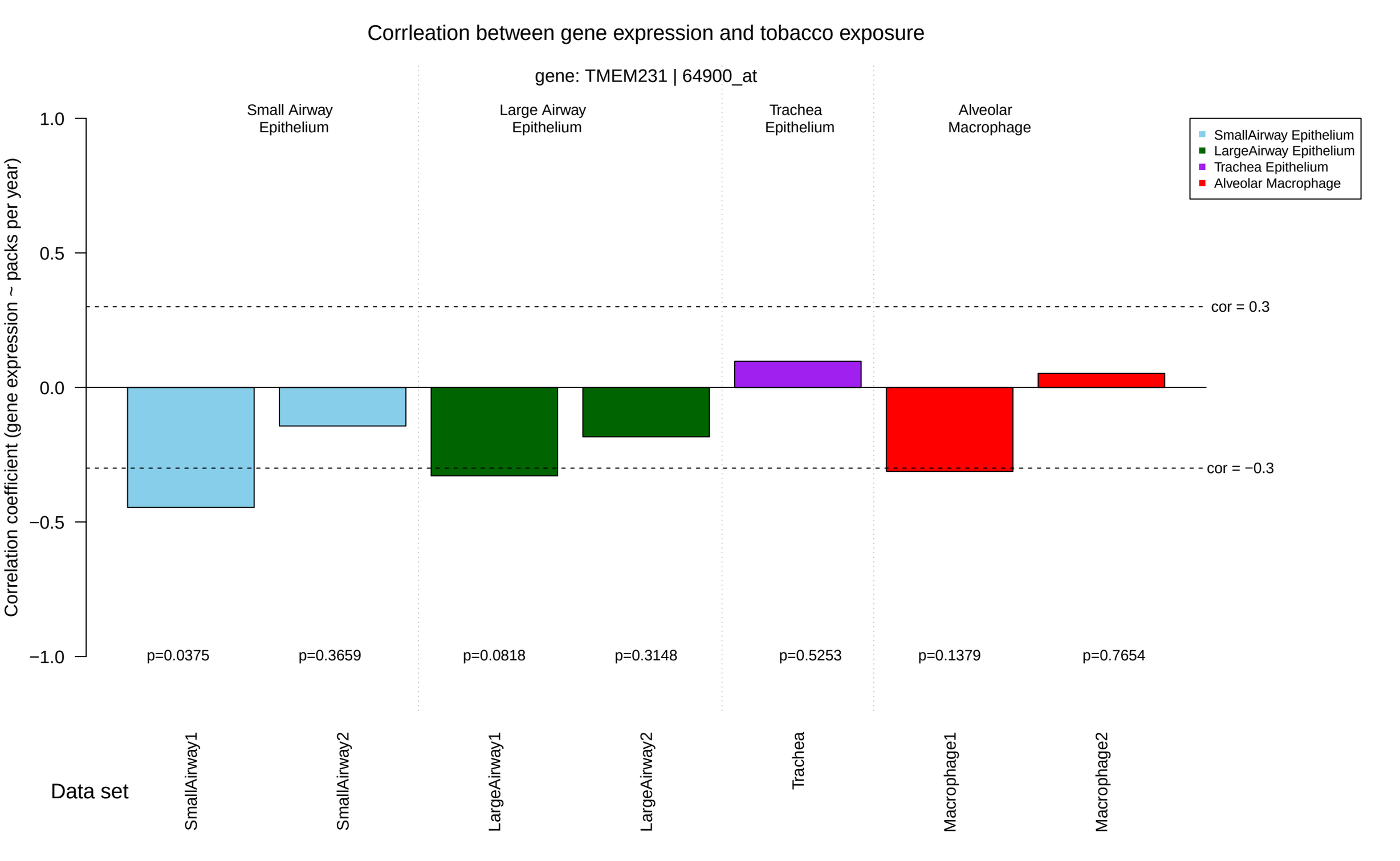

set: LargeAirway1 (p=0.0324)
TMEM231 | 64900_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3439)
TMEM231 | 64900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3916)
TMEM231 | 64900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0053)
TMEM231 | 64900_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9241)
TMEM231 | 64900_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM231 | 64900_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0375
p=0.3659
p=0.0818
p=0.3148
p=0.5253
p=0.1379
p=0.7654
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2